◆ Architecture Exercise 📐

AWS Solutions Architect Associate (SAA-C03)

Topic:

EC2 Instance Types • High Availability • Fault Tolerance • Scaling Concepts

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◈ Real SAA Exam Scenario

A company hosts a highly popular e-commerce website on AWS.

The company experiences unpredictable traffic spikes during special promotions and Black Friday sales.

Management requires the following:

✓ High Availability

✓ Fault Tolerance

✓ Automatic Scaling

✓ Cost Optimization

✓ Secure Architecture

As the AWS Solutions Architect, design a solution that meets these requirements.

◈ Step 1 — Understand the Requirements

Before selecting AWS services, identify what the business needs.

Requirement 1

Application must remain available if a server fails.

Requirement 2

Application must automatically handle traffic spikes.

Requirement 3

Application should avoid paying for unused resources.

Requirement 4

Database must remain available during failures.

Requirement 5

Architecture should follow AWS best practices.

◈ Step 2 — Select the Correct AWS Services

Compute Layer

Use:

- Amazon EC2

Purpose:

Run the application servers.

Load Balancing Layer

Use:

- Application Load Balancer (ALB)

Purpose:

Distribute incoming traffic across multiple EC2 instances.

Scaling Layer

Use:

- Auto Scaling Group (ASG)

Purpose:

Automatically launch or terminate EC2 instances based on demand.

Database Layer

Use:

- Amazon RDS Multi-AZ

Purpose:

Provide automatic database failover.

Networking Layer

Use:

- VPC
- Public Subnets
- Private Subnets
- Internet Gateway
- NAT Gateway

Purpose:

Create a secure network architecture.

◈ Step 3 — Recommended Architecture

```
                Internet
                   |
                   ▼

        Application Load Balancer
                 (ALB)

                   |
       ┌───────────┴───────────┐
       ▼                       ▼

   EC2 Instance A          EC2 Instance B
```

```
      AZ-A                     AZ-B
          └─────────────┬─────────────┘
                        │
              Auto Scaling Group
                        │
      ┌─────────────────┴─────────────────┐
      ▼                                   ▼
Private App Subnet A              Private App Subnet B
                        │
                        ▼
                  Amazon RDS
                  Multi-AZ
                        │
                 Standby Database
```

◈ High Availability Analysis

What is High Availability?

The ability of a system to remain available despite failures.

How Is High Availability Achieved?

Using:

- Multiple Availability Zones
- Load Balancers
- Auto Scaling
- Multi-AZ Databases

Example

If AZ-A experiences an outage:

```
AZ-A  ✕ Failed
```

`AZ-B ✓ Available`

The application continues running.

Users remain connected.

◈ Fault Tolerance Analysis

What is Fault Tolerance?

The ability to continue operating when components fail.

Example

If one EC2 instance crashes:

`EC2-A ✕ Failed`

`EC2-B ✓ Healthy`

The Application Load Balancer automatically routes traffic to healthy servers.

No manual intervention is required.

SAA Exam Tip

High Availability reduces downtime.

Fault Tolerance eliminates service interruption.

◈ Scaling Analysis

Vertical Scaling

Also known as:

`Scale Up`

Example:

```
t3.micro
```

↓

```
t3.large
```

More CPU and memory are added.

Horizontal Scaling

Also known as:

```
Scale Out
```

Example:

```
2 EC2 Instances
```

↓

```
10 EC2 Instances
```

More servers are added.

Which Scaling Method Does AWS Prefer?

AWS generally prefers:

✓ Horizontal Scaling

Because it improves:

- Availability
- Fault Tolerance
- Elasticity

◈ Instance Type Selection

General Purpose

Examples:

- T3
- T4g

- M5

Best For:

- Web Applications
- Development Environments

Compute Optimized

Examples:

- C5
- C6i

Best For:

- Scientific Computing
- Machine Learning
- CPU-intensive Workloads

Memory Optimized

Examples:

- R5
- R6i

Best For:

- Databases
- SAP HANA
- In-Memory Applications

Storage Optimized

Examples:

- I3
- D2

Best For:

- Big Data

- Data Warehouses

◈ SAA Exam Questions

Question 1

A company wants to distribute incoming traffic across multiple EC2 instances.

Which AWS service should be used?

A. Route 53

B. CloudFront

C. Application Load Balancer

D. Auto Scaling

Answer

✓ C. Application Load Balancer

Question 2

A company requires automatic database failover.

Which solution should be used?

A. Read Replica

B. EBS Snapshot

C. Multi-AZ RDS

D. Security Group

Answer

✓ C. Multi-AZ RDS

Question 3

A company increases an EC2 instance from t3.micro to t3.large.

What type of scaling is this?

A. Horizontal Scaling

B. Vertical Scaling

C. Auto Scaling

D. Elastic Scaling

Answer

✓ B. Vertical Scaling

Question 4

A company needs EC2 instances automatically added when CPU utilization exceeds 70%.

Which service should be used?

A. CloudWatch

B. Route 53

C. Auto Scaling Group

D. IAM

Answer

✓ C. Auto Scaling Group

Question 5

Which architecture provides the highest availability?

A. One EC2 instance in one Availability Zone

B. Two EC2 instances in one Availability Zone

C. Multiple EC2 instances across multiple Availability Zones

D. One large EC2 instance

Answer

✔ C. Multiple EC2 instances across multiple Availability Zones

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◈ SAA Exam Memory Tricks

Instance Types

Remember:

G-C-M-S

Letter	Meaning
G	General Purpose
C	Compute Optimized
M	Memory Optimized
S	Storage Optimized

Scaling

```
Scale Up   = Bigger Server

Scale Out  = More Servers
```

Availability

```
High Availability
=
Stay Online
```

Fault Tolerance

```
Fault Tolerance
=
Continue Operating After Failure
```

◈ Key Exam Takeaways

Before the exam, remember:

✔ General Purpose = Balanced Workloads

✓ Compute Optimized = CPU Intensive

✓ Memory Optimized = Database Workloads

✓ Storage Optimized = Big Data

✓ High Availability = Multiple AZs

✓ Fault Tolerance = Survive Failures

✓ Vertical Scaling = Scale Up

✓ Horizontal Scaling = Scale Out

✓ Auto Scaling = Automatic Capacity Adjustment

✓ Load Balancer = Traffic Distribution

✓ Multi-AZ RDS = Database Failover

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Exam Focus: If you understand why AWS recommends **Multi-AZ + Load Balancer + Auto Scaling**, you have mastered one of the most tested architecture patterns in the AWS Solutions Architect Associate exam.